

A 2.5D Approach to Training Robust 3D Pose Estimators in Specialized Exercise Settings

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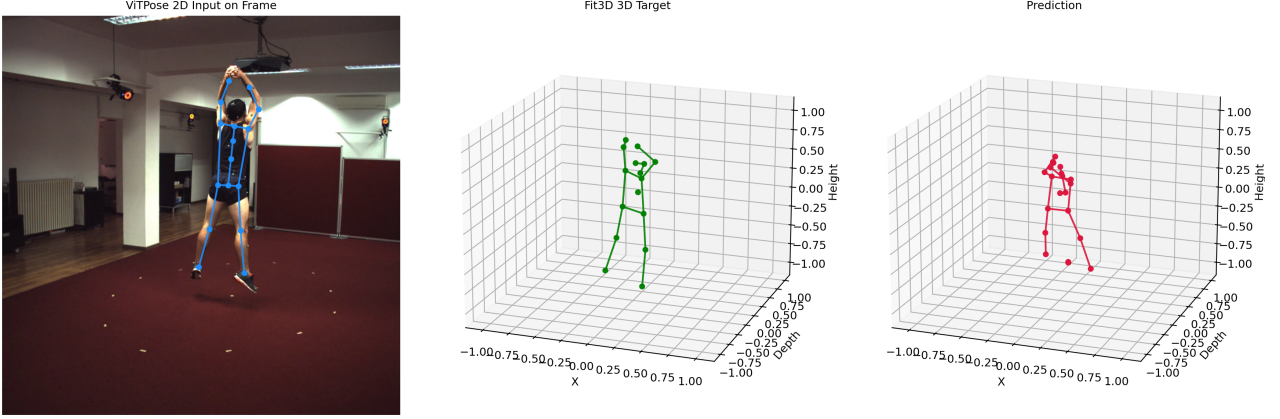


Figure 1: Fit3D Pose Estimation Task

Abstract

3D human pose estimation in specialized domains such as gym exercise is bottlenecked by the absence of 3D-labeled training data. We pursue three parallel hybrid 2.5D investigations of cross-domain adaptation. Two threads share a target-domain weak-supervision loop on different backbones — VideoPose3D fine-tuned with Detectron2-extracted Fit3D 2D pseudo-labels, and a frozen MotionBERT DSTformer with LoRA adapters and a knowledge-distillation regularizer. A third thread investigates whether cross-domain robustness can be replicated without any target-domain data, by substituting this target-domain for a diverse 2D substitute (COCO) and reconciling skeleton discrepancies through a learned skeleton-bridging autoencoder (ACAE). The MotionBERT thread reduces Fit3D MPJPE by **19.2%** (299→242 mm) at a 4 mm H36M cost, with a knowledge-distillation weight tracing a clean cross-domain Pareto frontier. The VideoPose3D thread cuts Fit3D MPJPE by **53%** (875→408 mm). The ACAE+COCO thread improves Fit3D MPJPE by **23.5%** without ever training pose estimators on Fit3D data. Pretrain quality, target-data availability, and skeleton compatibility emerge as three roughly orthogonal levers.

1 Introduction

3D pose estimation has matured rapidly in lab-captured settings, but this sterile setting is not generalizable to real-world contexts. We evaluate the gym data as one such opposite extreme where subjects find themselves in flexible poses and joints are often occluded from the camera. Our project asks: **can we adapt a 3D pose estimator to such a specialized domain using only weakly-supervised 2D signals on the target distribution, while preserving its lab-data accuracy — and what should we do when even target-domain 2D isn't available?**

The proposal laid out a hybrid 2.5D plan, but the previous midpoint milestone showed that naively adding 2D gym video to an H36M-trained baseline *worsened* in-domain accuracy (MPJPE on H36M increased from 47.3 to 51.7 mm). That negative result motivated three parallel responses, each holding a different variable fixed:

1. **Backbone strength.** Does a stronger pretrained init (MotionBERT DSTformer) tolerate 2D weak supervision better than a from-scratch ConvNet? We add LoRA + knowledge distillation to control drift from the pretrained anatomical prior.
2. **Cleaner target-domain 2D.** Was the milestone regression about supervision quality? We re-run the original VideoPose3D pipeline with Detectron2-extracted Fit3D 2D pseudo-labels and an optional ACAE preprocessing step.

3. **Target-data-free generalization.** What if no target-domain 2D is available? We test whether COCO 2D supervision routed through a learned ACAE skeleton bridge can still drive target-domain gains.

Success criteria. A given thread succeeds if it (a) reduces *raw* Fit3D MPJPE by a significant margin over the zero-shot version of its backbone, and (b) keeps in-domain H36M error from regressing, retaining learned priors.

Who benefits. Form-checking apps, physical therapy monitoring, and AR/VR fitness coaching all need 3D pose for exercises that lab datasets do not contain. Our finding is that LoRA-based hybrid fine-tuning provides such adaptation cheaply, and a single regularizer weight controls the trade-off between target-domain adaptation and source-domain preservation (Figure 2). The ACAE+COCO thread additionally points to a more generalized application where no target-domain data is even available. You can find the code here: <https://github.com/JasonLai150/2.5DPoseEstimator/tree/main>

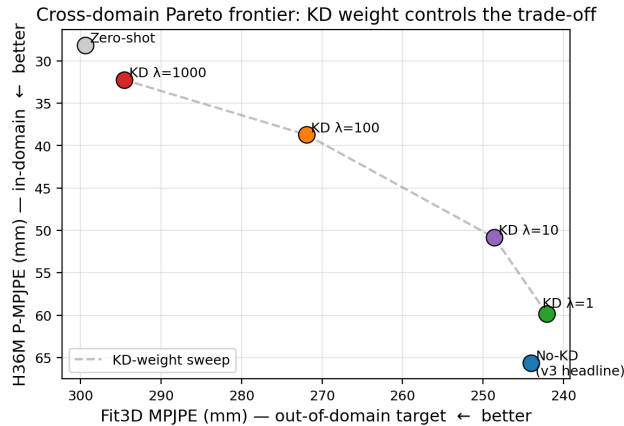


Figure 2: Cross-domain Pareto frontier traced by sweeping the knowledge-distillation weight λ_{kd} on the MotionBERT thread.

2 Problem Formulation and Related Work

Problem. Let $x \in \mathbb{R}^{T \times J \times 2}$ be a temporal sequence of 2D joint coordinates ($T = 243$ frames, $J = 17$ joints, normalized to $[-1, 1]$). The objective is to learn $f_\theta : \mathbb{R}^{T \times J \times 2} \rightarrow \mathbb{R}^{T \times J \times 3}$ producing root-relative 3D poses (in meters) in the camera frame. Training data comprises a 3D-labeled set $\mathcal{D}_{3D} = \{(x_i, y_i)\}$ from H36M and a 2D-only set $\mathcal{D}_{2D} = \{(x_j, c_j, K_j)\}$ from a target or surrogate domain (Fit3D videos for threads 1–2; COCO for thread 3), where c_j is the per-frame absolute pelvis camera position and K_j are camera intrinsics. Evaluation is on H36M-test (in-domain) and held-out Fit3D subjects (target-domain) under MPJPE / P-MPJPE / BLI.

Related work. Three lines are closely relevant to our final pipelines.

Convolutional temporal lifting. VideoPose3D (Pavlo et al., 2019) lifts 2D sequences to 3D with dilated temporal convolutions and introduces a back-projection consistency loss for semi-supervised training. It serves as the backbone for two of our three threads, and is a valid foundation given the abundance of video data in our target-domain.

Spatio-temporal transformers. MotionBERT (Zhu et al., 2023) introduces a dual-stream spatio-temporal transformer (DSTformer) pre-trained on AMASS via Masked Pose Modeling; APTPose (Yang et al., 2024) extends DSTformer with a reprojection module for hybrid supervision. Our MotionBERT thread builds on the public `MB_ft_h36m` checkpoint and adds LoRA adapters + knowledge distillation.

Skeleton bridging. Sárándi et al. (2023) propose an Affine Combining Autoencoder (ACAE) that reconciles differing skeleton conventions across datasets. We use ACAE as a preprocessing module in one thread and as a full skeleton bridge in another. 2D pseudo-labels come from Detectron2 (Wu et al., 2019) and ViTPose (Xu et al., 2022).

Table 1: Positioning relative to prior pipelines and across our three threads.

Method	Backbone	Weak supervision	Adapter / bridge	Tunable trade-off
VideoPose3D	TCN	reprojection	—	—
MotionBERT	DSTformer	pretrain only	—	—
APTPose	DSTformer	reprojection	—	—
Ours: VP3D + Fit3D-2D	VideoPose3D	Fit3D Detectron2 2D	optional ACAE-pre	data-subset
Ours: MB-LoRA-KD	DSTformer	Fit3D 2D + biomech	rank-2 LoRA	KD weight λ_{kd}
Ours: ACAE + COCO	VideoPose3D + ACAE	COCO ViTPose 2D	learned ACAE	data-subset

3 Method

3.1 Hypotheses (CS 7643)

H1 — Target-domain 2D supervision helps regardless of backbone. Adding target-domain 2D pseudo-labels under a hybrid 2.5D loop reduces Fit3D MPJPE relative to the zero-shot backbone, but the gain depends on backbone strength. *Expected:* both VP3D + Fit3D-2D and MB + LoRA pipeline will beat their zero-shot baseline on Fit3D, with the stronger pretrained backbone (MotionBERT) producing a lower absolute Fit3D MPJPE.

H2 — KD weight controls the cross-domain Pareto frontier. A frozen-teacher distillation term with weight λ_{kd} cleanly interpolates between “max Fit3D adaptation” (low λ_{kd}) and “max H36M preservation” (high λ_{kd}). *Expected:* sweeping $\lambda_{kd} \in \{1, 10, 100, 1000\}$ produces linear trade-offs, and the KD residual remains active throughout training (unlike biomechanical regularizers, which converge to inactive once anatomical validity is satisfied).

H3 — ACAE + diverse 2D substitute can adapt without target-domain data. When no target-domain 2D is available, a learned skeleton bridge (ACAE) can reconcile a diverse general-purpose 2D dataset (COCO) with the supervised 3D source. *Expected:* an H36M+Fit+COCO ACAE-bridged pipeline improves Fit3D performance over a target-data-free H36M+Fit baseline.

3.2 Shared Architecture and Composite Loss

All three threads optimize a variant of:

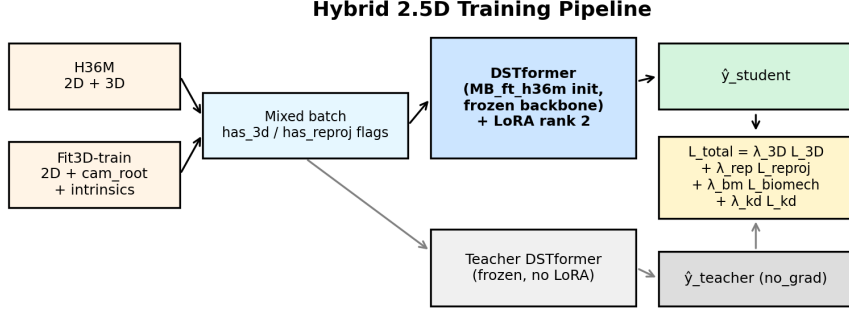
$$\mathcal{L} = \lambda_{3D} \mathcal{L}_{3D} + \lambda_{\text{reproj}} \mathcal{L}_{\text{reproj}} + \lambda_{\text{biomech}} \mathcal{L}_{\text{biomech}} + \lambda_{kd} \mathcal{L}_{kd} \quad (1)$$

with per-sample gating skipping $\mathcal{L}_{3D}$ / $\mathcal{L}_{kd}$ on weakly-supervised samples and $\mathcal{L}_{\text{reproj}}$ on 3D-supervised samples. Components: $\mathcal{L}_{3D}$ is MPJPE on H36M-supervised samples; $\mathcal{L}_{\text{reproj}}$ is L1 between input 2D and projected 3D (MotionBERT thread recovers absolute camera-space pose by adding the saved per-frame pelvis position before projecting, $\hat{y}_{j,t}^{\text{abs}} = \hat{y}_{j,t} + c_{j,t}$); $\mathcal{L}_{\text{biomech}}$ is bilateral-symmetry plus an anatomical-hinge term ($\text{ReLU}(\theta - 160^\circ)$); $\mathcal{L}_{kd}$ (MotionBERT thread only) is MSE between student and a frozen MB_ft_h36m teacher on H36M samples.

3.3 Backbone Variants

VideoPose3D + Fit3D-2D: A VideoPose3D temporal convolutional model fine-tuned for 10 epochs from the H36M-only epoch-80 checkpoint. Fit3D RGB videos are processed with Detectron2 to extract 2D keypoints, converted to VideoPose3D’s .npz format, and added as an unlabeled 2D branch. Two data settings are evaluated: a *partial* set (s03/s04/s05) and a *larger* set (s03/s04/s05/s07). The larger setting contains more than twice as many processed Fit3D videos as the partial setting, allowing us to test whether simply increasing the amount of unlabeled target-domain 2D data improves adaptation.

MotionBERT + LoRA + KD: A 5-block DSTformer loaded from the public MB_ft_h36m.bin and frozen. LoRA adapters (Hu et al., 2021) at rank 2 are attached to the qkv and proj linear layers of every attention block, giving 122,880 trainable parameters (0.29% of total). The Fit3D 2D branch uses *projected ground-truth*



Notes: L_{3D} and L_{kd} gated to $has_3d=True$ samples. L_{reproj} recovers absolute 3D via $pred + cam_root$, projects with per-sample (fx, fy, cx, cy) . $L_{biomech}$ (symmetry + hinge) always active.

Figure 3: Training pipeline (MotionBERT thread shown). The VideoPose3D thread replaces the backbone block; the ACAE thread additionally wraps it with a learned skeleton bridge.

3D for clean supervision. A KD-regularized variant adds a frozen MB_ft_h36m teacher pulling the student back toward its pretrained behavior on H36M samples; sweeping $\lambda_{kd} \in \{1, 10, 100, 1000\}$ produces a Pareto frontier.

ACAE + COCO substitute: A VideoPose3D model wrapped in a pretrained ACAE skeleton bridge trained on the same datasets being evaluated: inputs are mapped into the unified ACAE skeleton, encoded into a 17-joint latent, passed through VideoPose3D, decoded back, and loss is computed. A pretrained ViTPose produces 2D keypoints as input for the autoencoder. Two checkpoints are compared: an H36+Fit baseline (3D + biomech only) and an H36+Fit+COCO setting that adds up to 100,000 COCO 2D ViTPose detections with an additional 2d reprojection loss term, while retaining some H36 data as well to prevent forgetting. Unlike the above experiments, the pose estimators themselves do not see Fit3D 2D videos during training, though the autoencoder has to in order to create an accurate latent space.

3.4 Training, Evaluation, Reproducibility

Training. All three threads use AdamW. MotionBERT thread: $lr=10^{-4}$, weight decay 0.01, 5-epoch warmup + cosine decay to 10^{-6} , batch 16, sequence length 243, 15 epochs. VideoPose3D thread: 10 fine-tuning epochs from the H36M epoch-80 checkpoint with the standard VideoPose3D schedule. ACAE thread: $lr=10^{-4}$, sample stride 2, 20 epochs, ACAE frozen. All checkpoints selected by Fit3D val MPJPE.

Evaluation. Metrics differ slightly by thread. The VideoPose3D thread reports H36M MPJPE, P-MPJPE, N-MPJPE, and MPJVE, plus Fit3D root-relative MPJPE. The other threads report MPJPE, P-MPJPE, and BLI on Fit3D-s11 and H36M-test. Skeleton mappings differ across threads (manual bone-ratio for MotionBERT, naive truncation for VideoPose3D, learned ACAE for the third), so absolute MPJPE numbers are not strictly comparable across threads — only the relative gain over each thread’s own baseline is.

Hardware. PyTorch 2.6 + CUDA 11.8 on Georgia Tech PACE-ICE; runs scheduled to L40S/A40/A100/H100/H200 nodes.

4 Data

- **Human3.6M** (Ionescu et al., 2014): 3.6 M lab-captured frames, S1/5/6/7/8 train, S9/11 test. The MotionBERT thread windows at length 243, stride 243 (1,529 train / 532 test windows).
- **Fit3D** (Fieraru et al., 2021): 47 fitness actions across 11 subjects, with calibrated cameras and 3D ground truth. The MotionBERT thread projects ground truth 3D to 2D itself; the VideoPose3D thread uses

Detectron2 on RGB to obtain 2D keypoints; the ACAE thread uses ViTPose. For the VideoPose3D thread, the partial adaptation setting uses Fit3D subjects s03/s04/s05, while the larger setting uses s03/s04/s05/s07; evaluation is performed on held-out Fit3D videos processed separately from these adaptation subjects.

- **COCO 2017** (Lin et al., 2014): 250K images of in-the-wild 2D-annotated people. Used only by the ACAE thread (up to 100K pose keypoints extracted via ViTPose).

5 Experiments and Results

5.1 Baselines

Table 2: Zero-shot baselines on H36M-test and Fit3D.

Baseline	H36M MPJPE	H36M P-MPJPE	Fit3D MPJPE	Fit3D P-MPJPE	Fit3D BLI
VideoPose3D H36M-only (epoch 80)	47.3	36.9	874.7	160.9	—
MotionBERT MB_ft_h36m zero-shot	520.8*	28.2	299.4	163.3	0.0037

*Reflects a known scale mismatch between MotionBERT outputs and our processed H36M targets; published P-MPJPE is 37.2 mm and our 28.2 mm reproduction matches.

The two backbones produce wildly different starting points on Fit3D — 875 mm vs 299 mm — so backbone choice alone changes the cross-domain gap by nearly $3\times$. This validates the findings from the last milestone report: a from-scratch ConvNet has no anatomical prior to fall back on outside H36M.

5.2 Hybrid 2.5D with Target-Domain 2D

Both the VideoPose3D thread and MotionBERT thread fine-tune their respective backbones with target-domain Fit3D 2D supervision. We report them together because they answer the same question — *does adding target-domain 2D help?* — at different points on the backbone-strength axis.

Table 3: VideoPose3D + Fit3D-2D adaptation. Initialized from H36M epoch 80; 10 additional epochs. Fit3D MPJPE uses the local naive `joints3d_25[:, :17]` mapping, so these values are intended for within-thread comparison.

Run	H36M MPJPE	H36M P-MPJPE	H36M N-MPJPE	H36M MPJVE	Fit3D MPJPE
H36M baseline	47.3	36.9	45.3	2.82	874.7
+ Fit3D-2D partial (s03/s04/s05)	49.7	38.5	47.7	2.80	407.7
+ Fit3D-2D + ACAE partial (s03/s04/s05)	49.5	38.0	47.0	2.84	560.4
+ Fit3D-2D larger (s03/s04/s05/s07)	49.6	37.8	47.2	2.83	736.8
+ Fit3D-2D + ACAE larger (s03/s04/s05/s07)	49.1	38.1	46.9	2.81	544.6

The direct partial adaptation cuts Fit3D MPJPE by **53%** ($874.7 \rightarrow 407.7$ mm) at a 2.4 mm H36M MPJPE cost, making it the strongest VideoPose3D result. The ACAE-preprocessed partial model is worse than the direct partial model ($407.7 \rightarrow 560.4$ mm), suggesting that the extra skeleton-space transformation can distort useful pose information when the direct pseudo-label set is already clean and well matched to the VideoPose3D input distribution. However, ACAE helps in the larger s03/s04/s05/s07 setting, improving Fit3D MPJPE from 736.8 mm to 544.6 mm. Overall, ACAE is best interpreted as a regularizing skeleton-harmonization tool that stabilizes noisier or more heterogeneous unlabeled data, not as a guaranteed improvement over direct 2D adaptation.

The best MotionBERT run reduces Fit3D MPJPE by **19.2%** ($299.4 \rightarrow 242.1$ mm) at +6% Fit3D P-MPJPE and an H36M P-MPJPE shift of $28 \rightarrow 60$ mm. Sweeping λ_{kd} from 1 to 1000 traces the Pareto frontier in Figure 2: at $\lambda=1000$, H36M P-MPJPE recovers to 32.3 mm (within 4 mm of zero-shot) at the cost of giving back most of the Fit3D gain. The KD residual term shrinks $20\times$ across the sweep ($0.0020 \rightarrow 0.0001$), confirming the trade-off is real pressure being applied. Figure 4 visualizes the full ablation.

Cross-thread reading. Both backbones validate H1: target-domain 2D supervision improves Fit3D, with mild H36M cost. The MotionBERT thread reaches better numbers on both axes — pretrained anatomical priors

Table 4: MotionBERT + LoRA + KD ablation. All runs init MB_ft_h36m, 15 epochs, AdamW lr = 10^{-4} , batch 16. λ defaults: $\lambda_{3D} = 1.0$, $\lambda_{\text{reproj}} = 0.5$, $\lambda_{\text{biomech}} = 1.0$, $\lambda_{kd} = 0$. Run-specific changes in *italics*.

Run	H36M MPJPE	H36M P-MPJPE	Fit3D MPJPE	Fit3D P-MPJPE	Fit3D BLI
MotionBERT zero-shot	520.8	28.2	299.4	163.3	0.0037
v3 + low LR (<i>lr=1e-5</i>)	520.8	28.2	302.3	166.1	0.0037
v3 + biomech $\times 2$	520.8	28.2	302.3	166.1	0.0037
v3 + no H36M ($\lambda_{3D} = 0$)	546.8	69.5	259.2	181.5	0.0061
v3 + rank-2 LoRA	215.2	170.7	260.2	185.0	0.0336
v3 + no H36M + rank-2	550.2	65.7	244.0	173.8	0.0054
v4 + KD $\lambda = 1$ (best)	536.3	59.8	242.1	173.1	0.0049
v4 + KD $\lambda = 10$	527.3	50.9	248.6	175.5	0.0056
v4 + KD $\lambda = 100$	523.1	38.7	271.9	176.5	0.0044
v4 + KD $\lambda = 1000$	522.7	32.3	294.5	178.4	0.0036

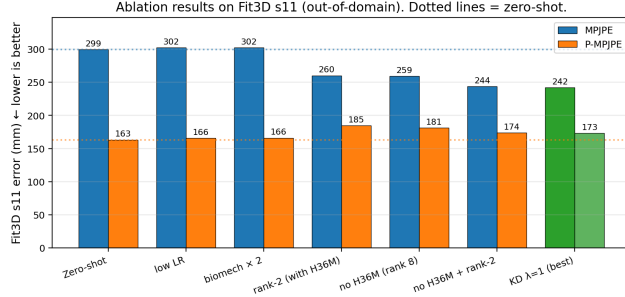


Figure 4: MotionBERT thread: ablation comparison on Fit3D-s11 MPJPE / P-MPJPE.

carry most of the load. Both threads also show that more target-domain 2D is not automatically better, and in each case a structure-preserving counterweight (ACAE preprocessing for VideoPose3D; KD for MotionBERT) was the missing piece.

5.3 Per-Action Analysis

For the headline MotionBERT run (Figure 5), the eight hardest actions (P-MPJPE > 220 mm) are all floor-based, inverted, or heavily occluded: pushup (264), burpees (255), diamond_pushup (254), warmup_1 (254), mule_kick (233), man_maker (229), warmup_5 (224), warmup_19 (218). The eight easiest (< 130 mm) are all standing dumbbell exercises that match H36M’s body-orientation distribution closely. The domain gap correlates strongly with deviation from H36M’s standing/sitting prior — pointing to where targeted data collection would have the largest effect.

5.4 Target-Data-Free Generalization (ACAE + COCO)

When target-domain 2D is unavailable, can a diverse 2D substitute (COCO) routed through a learned skeleton bridge still help? Table 5 reports two ACAE-bridged checkpoints.

Table 5: ACAE + ViTPose pipeline. Initialized from H36M VideoPose3D + frozen ACAE; 20 epochs.

Run	H36M MPJPE	H36M P-MPJPE
Fit3D MPJPE	Fit3D P-MPJPE	Fit3D BLI
H36+Fit ACAE baseline (3D + biomech only)	83.82	58.55
406.00	205.45	56.16
+ COCO ViTPose 2D reprojection	197.04	310.71
310.71	164.62	13.74

Adding COCO improves Fit3D MPJPE by **23.5%** (406 $\rightarrow$ 311 mm) without the pose estimators being trained on Fit3D data. However, the degradation in H36M performance is significant (+113 mm) — but it does show

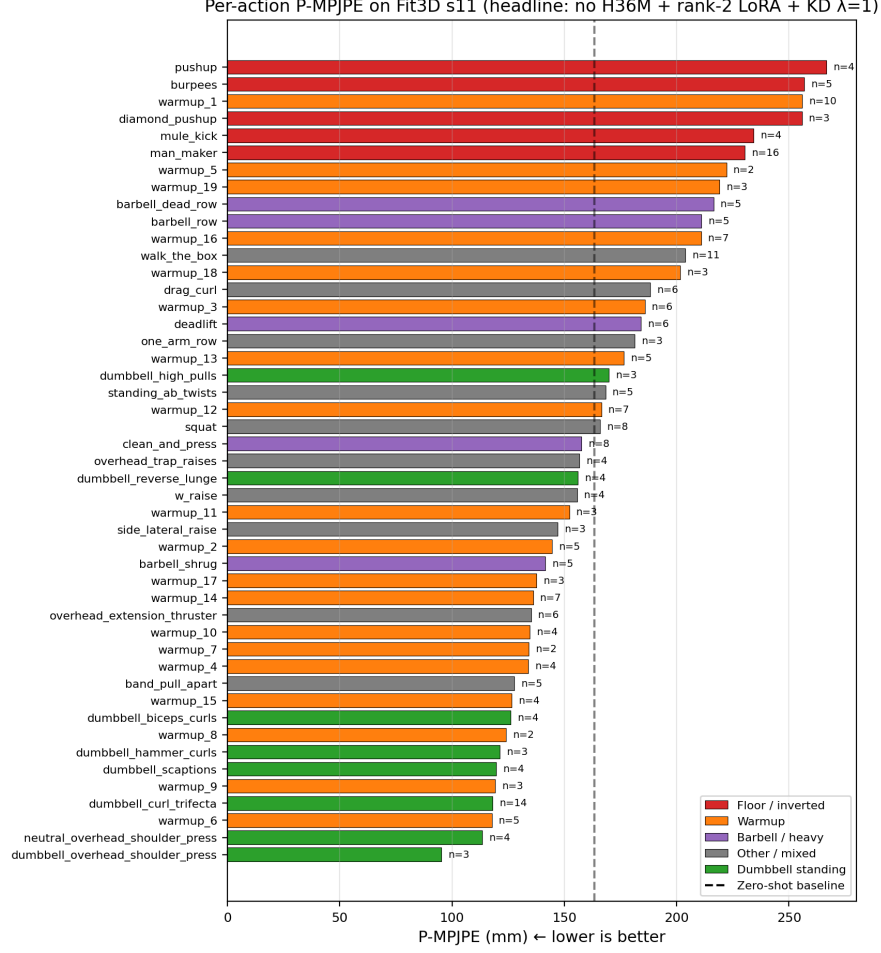


Figure 5: Per-action P-MPJPE on Fit3D-s11 for the headline MotionBERT run ($v4 + KD \lambda = 1$).

that diverse 2D + a learned skeleton bridge can drive cross-domain gains in a closer to *zero-target-data* pipeline. This shift in H36 performance could be attributed to the much noisier nature of COCO’s domain, and that choosing a more moderate distribution could still improve target-domain performance while sacrificing less H36 performance.

5.5 Hypothesis Confirmation

H1 (target-domain 2D helps; backbone matters): Confirmed. VideoPose3D thread -53% Fit3D, MotionBERT thread -19.2% Fit3D. The MotionBERT thread reaches lower absolute Fit3D MPJPE *and* better H36M P-MPJPE under KD, consistent with the prediction.

H2 (KD weight controls Pareto frontier): Confirmed. Sweeping λ_{kd} moves H36M P-MPJPE down from $59.8 \rightarrow 32.3$ mm and Fit3D MPJPE up from $242.1 \rightarrow 294.5$ mm. KD stays active throughout training; biomech hinge collapses to ≤ 0.02 within four epochs, which is why scaling $\lambda_{biomech}$ alone does nothing.

H3 (ACAE + COCO substitute can help without target data): Confirmed, partially. COCO improves Fit3D MPJPE 23.5% over the target-data-free baseline. The $+113$ mm H36M regression means this is not deployable as-is, and reducing the H36M loss is the natural next step.

5.6 Failure Modes

Naive hybrid training diverges. A v1 MotionBERT run with $\lambda_{\text{biomech}} = 0.1$, $\lambda_{\text{reproj}} = 0$ saw val P-MPJPE balloon from 0.28 to 0.49 m as $\mathcal{L}_{3D}$ collapsed — reproducing the milestone’s negative result on a new backbone, so the failure is independent of architecture.

LoRA at rank 2: scale fixed, structure broken. The “rank-2 alone” run (with H36M supervision) cuts H36M raw MPJPE 60% ($520.8 \rightarrow 215.2$) but blows up H36M P-MPJPE $6\times$ — at this capacity LoRA is mostly fixing MotionBERT’s known scale mismatch, not learning new joint geometry. Stronger KD recovers structure (BLI 0.0036 at $\lambda_{kd} = 1000$, matching zero-shot), confirming KD as a structure-preserving regularizer alongside its in-domain preservation role.

6 Conclusion

The proposal’s hybrid-supervision sketch is essentially what we ended up running, but two of its components turned out to be load-bearing in unexpected ways. The choice of pretrained init mattered far more than the loss formulation — MotionBERT alone closed most of the Fit3D gap before any fine-tuning. And the biomechanical regularizer — central to the proposal — converges to inactive within four epochs and is *not* what stops divergence in practice. The actual stabilizers were knowledge distillation in the MotionBERT thread and, in some settings, ACAE preprocessing in the VideoPose3D thread. ACAE improved the larger and noisier VideoPose3D setting, but worsened the cleaner partial setting, so it is better understood as a conditional regularizer rather than a universally beneficial preprocessing step. The ACAE in the COCO thread was also shown to stabilize performance against an unseen distribution, while its substantial cost to H36 performance could be mitigated with a more conservative diverse 2D dataset.

Limitations. (1) Different Fit3D skeleton mappings across threads make absolute MPJPE numbers not strictly comparable — only the relative gain over each thread’s own baseline is. (2) Reprojection in the MotionBERT thread assumes ground-truth Fit3D intrinsics. (3) Floor-based exercises (pushups, burpees) remain over 250 mm P-MPJPE in all three pipelines — the LoRA capacity ceiling at rank 2 is hit there. (4) Fit3D is cleaner than true in-the-wild gym video. (5) The ACAE+COCO thread’s H36M regression is too large for as-is deployment, and the pipeline does still expect access to the target distribution to train just the ACAE.

Future work. (i) **Combine the threads.** A MotionBERT + ACAE + COCO pipeline would plausibly reach Fit3D MPJPE comparable to this paper’s MotionBERT thread without needing target-domain data. (ii) **KD curriculum.** Annealing λ_{kd} over training (start strong, decay late) would let the model use the anchor only when it most needs it. (iii) **Per-action LoRA adapters** routed by detected pose family ($47 \times 122k \approx 5.7$ M params) would isolate the floor-pose failure mode. (iv) **Confidence-weighted reprojection** in the VideoPose3D thread would down-weight noisy Detectron2 frames.

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Team Contributions

Member	Contributions
Candice Chen	VideoPose3D + Fit3D-2D adaptation pipeline; Detectron2 keypoint extraction and conversion to VideoPose3D .npz format; H36M VideoPose3D baseline training from the epoch-80 checkpoint and Fit3D-2D fine-tuning runs (partial and larger settings); ACAE-preprocessing variants for both VideoPose3D adaptation settings; H36M and Fit3D MPJPE evaluation tables; analysis of the data-quality, data-volume, and skeleton-harmonization tradeoffs.
Alec Cheng	ACAE skeleton-bridging pipeline integrated end-to-end with VideoPose3D; ViTPose 2D extraction on COCO and Fit3D; camera-aware Fit3D evaluation using per-video calibration; H36+Fit ACAE baseline and H36+Fit+COCO bridge training; inference and analysis of target-data-free generalization.
Jason Lai	MotionBERT + LoRA + KD pipeline; reprojection loss with per-frame absolute-depth recovery; IMAR coordinate-convention fix; principled bone-ratio Fit3D 25→17 skeleton mapping; LoRA rank ablation; KD-weight Pareto sweep ($\lambda_{kd} \in \{1, 10, 100, 1000\}$); per-action P-MPJPE analysis on Fit3D-s11; report writing and figures.